

27. MECHANISM OF TRANSDUCTION

DURATION : 08:39

STUDY SHEET

COURSE SUMMARY

This video explores in depth the mechanism of visual transduction, detailing the path of light through the different layers of the retina, from the photoreceptors to the optic nerve. The various cells of the retina, including cones, rods, bipolar cells, and ganglion cells, are examined to understand their role in visual information processing. The chemical transformation of rhodopsin and iodopsin upon light absorption, as well as the importance of vitamin A, are also discussed, highlighting the consequences of a deficiency on vision.

MECHANISM OF TRANSDUCTION

Light rays first pass through all layers of the retina to reach the **photoreceptors**. These photoreceptors are active due to their location in an area called the **pigment layer**, which constitutes the first layer. Photoreceptors are divided into two types: **cones** and **rods**.

After passing through this first layer, light reaches the **bipolar cells**, then the **ganglion cells**. The latter group their axons to form the **optic nerve**, where transduction occurs.

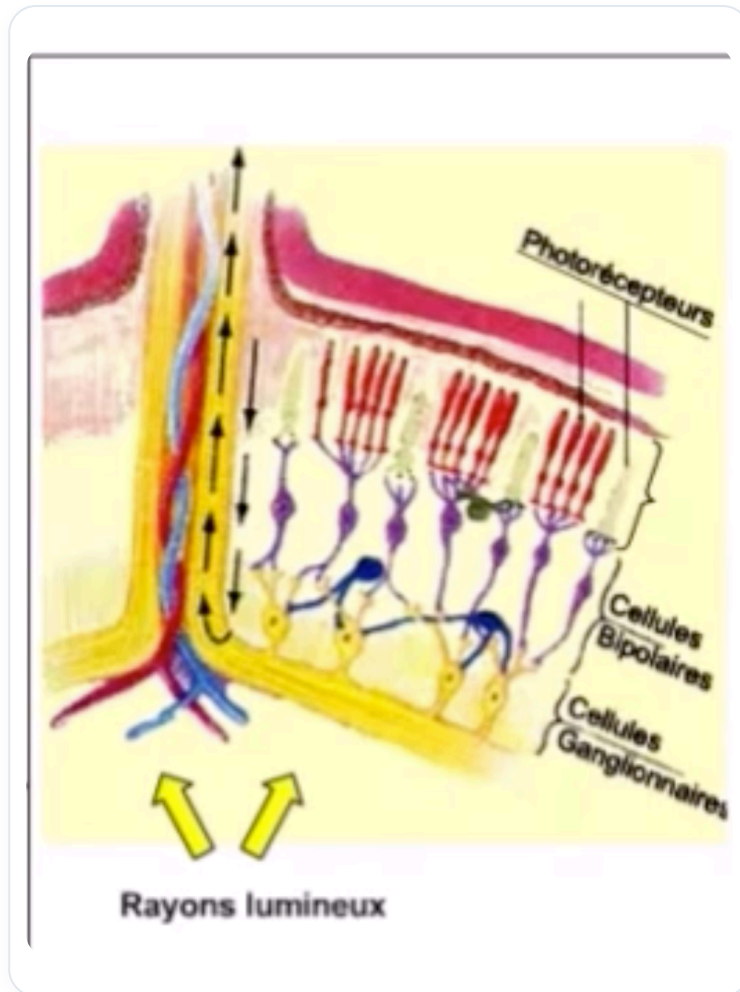


FIG. — Light pathway to the optic nerve

Delving deeper, we also observe **horizontal cells** that capture and redistribute information to the bipolar cells. Below these are the **amacrine cells**, which gather information before directing it to the ganglion cells. This complex system allows for the processing of a large amount of visual information.

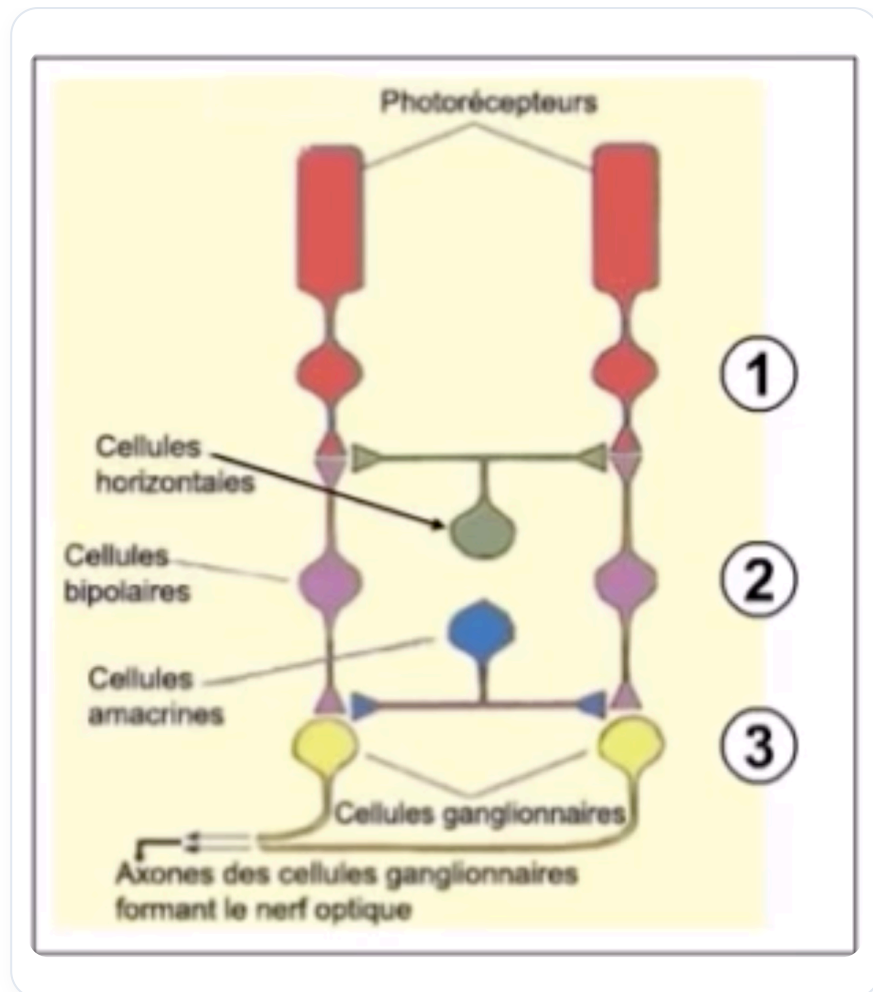


FIG. — Role of horizontal and amacrine cells

Let's take the example of a cone photoreceptor. It consists of an **outer segment**, an **inner segment**, a **nucleus**, and a **synaptic terminal**. In the outer segment, we find **plasma membranes** and a stack of small discs containing a protein called **opsin**. For rods, this opsin is called **rhodopsin**, while for cones, it is called **iodopsin**.

In rods, a small molecule called **retinal** is attached to rhodopsin. When light strikes these discs, retinal changes shape, transforming rhodopsin into **metarhodopsin**. This transformation triggers a series of chemical cascades at the cell membrane, preventing sodium entry and creating an ionic change.

This change in membrane permeability leads to a **membrane potential** that generates a **nerve impulse**. Thus, light, by passing through the layers, activates a pigment, beta-carotene, present in vitamin A, essential for activating rhodopsin and iodopsin.

Rods, located mainly in the peripheral retina, provide low-detail vision but are very sensitive to light, without colour perception. In contrast, cones offer detailed and coloured central vision, with specific types for blue, red, and green, corresponding to different wavelengths.

Rods consist of three parts: the outer segment, the inner segment, and the synaptic terminal. The outer segment contains small opsin discs, which, in the case of rods, is rhodopsin.

When light reaches these discs, rhodopsin transforms into metarhodopsin, leading to a chemical cascade that modifies the membrane's polarity, making it impermeable to sodium. This change creates an electrical current, essential for the transmission of visual information.

The **transduction** of light energy occurs in the photoreceptors through the absorption of light by opsin, which changes shape according to the light received. This process is crucial for vision, and vitamin A plays a fundamental role in retinal synthesis. A deficiency in vitamin A can lead to vision loss, particularly night blindness.